

Everton de Matos, Ph.D.

Home address: Abu Dhabi, United Arab Emirates

Linkedin: <https://www.linkedin.com/in/evermatos/>

Email: evermatos93@gmail.com

Nationality: Brazilian

Mobile : +971-58-590-2832

SUMMARY

- **Lead Security Research Engineer** at *Technology Innovation Institute*, United Arab Emirates, in the Cryptography Research Center department. Experience in virtualization for *Embedded Systems*, AI for Edge Devices, research on *seL4 Microkernel* for both *ARM* and *RISC-V* platforms, drivers, and kernel subsystems.
- **Ph.D. in Computer Science** from the Pontifical Catholic University of Rio Grande do Sul (PUCRS). Awarded a *Fulbright* scholarship to develop part of the doctoral dissertation at the University of Southern California (USC). Has several publications in important conferences and journals. Areas of interest are *embedded systems*, *information security*, *virtualization*, and *operating systems*.

EXPERIENCE

- **Technology Innovation Institute (5.5+ years)** Abu Dhabi, United Arab Emirates
 - Lead Security Research Engineer* *Apr 2023 - Current*
 - Senior Security Research Engineer* *Aug 2020 - Mar 2023*
 - **Research and Development:**
 - Research in Embedded Systems security and confidential computing for both ARM and RISC-V platforms.
 - Implementation and deployment of virtualization solutions using seL4 microkernel and hypervisor on ARM platform.
 - Bring-ups, OS support, feature set support, device virtualization, kernel modules.
 - Contributor to the Ghaf Virtualization Platform on the following areas: multi-VM logging, security policies, inter-VM transport layer, and VM and workload attestation.
 - Deployment of an AI-powered Intrusion Detection System (IDS) on an Edge Device within the Ghaf virtualization framework, optimizing real-time anomaly detection in constrained environments.
 - Proof-of-concept development-centered.
 - Mentoring of PhD students on their research activities.
 - Supervise projects sponsored by the company at highly prestigious universities around the world (e.g., UNSW, Purdue, Imperial College, PUCRS, and UMINHO).
 - Write scientific papers and technical reports, and organize and present research outcomes to higher management and a broader external audience.
- **ATITUS (2.5 years)** Passo Fundo, Brazil
 - Adjunct Professor* *Feb 2018 - Aug 2020*
 - **Teaching and Research:**
 - Taught undergraduate Computer Science courses, prepared lectures and assessments, and mentored students on academic and technical projects.
 - Courses taught included: Operating Systems, Computer Security, Internet of Things, Computer Architecture, and Robotics.
 - Supervised independent and group research projects, supported student presentations, and collaborated with faculty on research initiatives.
- **University of Southern California (1 year)** Los Angeles, CA
 - Visiting Researcher* *Aug 2018 - May 2019*
 - **Research:**
 - Awarded a Doctoral Dissertation Research Award (DDRA) grant from the Fulbright Brazil Commission.
 - Conducted research on Internet of Things (IoT), blockchain-based IoT marketplaces, and context-driven decision-making.
 - Authored scientific papers and technical reports based on the research outcomes.
- **Pontifical Catholic University of Rio Grande do Sul (6 years)** Porto Alegre, Brazil
 - Researcher* *Mar 2014 - Mar 2020*
 - **Research:**
 - Conducted research in embedded systems, focusing on context-awareness and security aspects of the Internet of Things.
 - Investigated context sharing as a mechanism to support a common definition of context information among heterogeneous IoT entities.
 - Designed and developed a context-sharing framework for IoT environments.
 - Implemented and validated a system to address research challenges related to context-awareness in IoT.
 - Authored scientific papers and technical reports based on the research outcomes.

EDUCATION

- **Pontifical Catholic University of Rio Grande do Sul** Porto Alegre, Brazil
PhD in Computer Science Mar 2016 - Mar 2020
Doctorate's dissertation: "Edge-centric context sharing architecture for the internet of things: context interoperability and context-aware security"
- **Pontifical Catholic University of Rio Grande do Sul** Porto Alegre, Brazil
MSc in Computer Science Mar 2014 - Mar 2016
Master's thesis: "Context-aware information services provision for IoT environments".
- **University of Passo Fundo** Passo Fundo, Brazil
BSc in Computer Science Feb 2010 - Jan 2014
Final project: "Development of a low-cost prototype to measure body balance".

SKILLS SUMMARY

- **Languages:** C, C++, Unix scripting, Python, Assembly, Java
- **Tools:** seL4, KVM, Docker, Linux, U-Boot, GIT, JIRA, Confluence
- **Platforms:** ARM, RISC-V

AWARDS AND HONORS

- **2020:** Approved with honor at the PhD in Computer Science - Pontifical Catholic University of Rio Grande do Sul
- **2018:** Doctoral Dissertation Research Award Scholarship - Fulbright
- **2016:** Approved with honor at the MSc in Computer Science - Pontifical Catholic University of Rio Grande do Sul
- **2015:** Finalist (among 10 best) in the contest of best Masters Thesis in Computer Science of Brazilian Computer Society

PUBLICATIONS

- The complete list of publications is available at [Google Scholar](#). The publications considering the last 3 years:
- **Journals:**
 - Arduino-based pedagogical aid for enhanced dexterity skill development in periodontal treatment training. *BMJ Innovations* 12 (1), 62-66, 2026. <https://doi.org/10.1136/bmjinnov-2024-001349>
 - Towards seL4 for Enhanced System Isolation and Security on Embedded Devices. *IEEE Open Journal of the Computer Society*, vol. 6, pp. 1329-1340, 2025. <https://doi.org/10.1109/OJCS.2025.3592377>
 - An Energy-Efficient Intrusion Detection Offloading Based on DNN for Edge Computing. *IEEE Internet of Things Journal*, vol. 12, no. 12, pp. 20326-20342, June 15, 2025. <https://doi.org/10.1109/IIOT.2025.3544060>
- **Conferences:**
 - Toward an Intrusion Detection System for a Virtualization Framework in Edge Computing (*Accepted*). In 2026 IEEE Wireless Communications and Networking Conference (WCNC).
 - An seL4-based Trusted Execution Environment on RISC-V. In 2024 International Wireless Communications and Mobile Computing (IWCMC), Ayia Napa, Cyprus, 2024, pp. 712-717. <https://doi.org/10.1109/IWCMC61514.2024.10592332>
 - Toward a Reliable Network-Based Intrusion Detection Model for SCADA: A Classification with Reject Option Approach. In 2024 International Joint Conference on Neural Networks (IJCNN), Yokohama, Japan, 2024, pp. 1-8. <https://doi.org/10.1109/IJCNN60899.2024.10650735>
 - A dynamic network-based intrusion detection model for industrial control systems. In 2023 IEEE 22nd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), Exeter, United Kingdom, 2023, pp. 1496-150. <https://doi.org/10.1109/TrustCom60117.2023.00204>
 - Integrating VirtIO and QEMU on seL4 for Enhanced Devices Virtualization Support. In IEEE 22nd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), Exeter, United Kingdom, 2023, pp. 1076-1085. <https://doi.org/10.1109/TrustCom60117.2023.00149>
 - Context-Aware Security in the Internet of Things: A Review. In 2020 37th International Conference on Advanced Information Networking and Applications (AINA-2023) (pp. 518-531). Springer. https://doi.org/10.1007/978-3-031-28694-0_49
 - A Dynamic Machine Learning Scheme for Reliable Network-Based Intrusion Detection. In 2020 37th International Conference on Advanced Information Networking and Applications (AINA-2023) (pp. 439-451). Springer. https://doi.org/10.1007/978-3-031-28451-9_39